Jordan Nguyen

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EDUCATION

UC Santa Cruz — *B.S. in Robotics Engineering | GPA: 3.83* *Anticipated Graduation: June 2027*

Relevant Coursework: Intro to Mechatronics, Microcontroller Design, Feedback Control Systems, Signals & Systems

Affiliations / Honors: College Track Oakland Scholar, Dean's List (2023-26), Oakland Promise Great Expectations Scholar (2023)

EXPERIENCE

Tactile Manipulation Lab **Santa Cruz, CA**

Undergraduate Research Assistant *January 2025 - Present*

- Designed and fabricated a ZED 2i camera mount for a Franka robotic arm using Onshape and 3D printing to provide a view of the gripper during manipulation, designing the interface to fit the Franka flange.
- Fabricated CoinFT capacitive force/torque sensors through silicone molding, PCB bonding, SMD soldering, and electrical assembly, troubleshooting 3–4 failed bonding attempts before producing functional sensor assemblies.
- Optimized resin-printed CoinFT molds through ~15 iterations, tuning part orientation, supports, and exposure time to reduce warping.
- Collected multi-axis sensor data using an ATI force/torque sensor and calibrated CoinFT using existing Python data-processing and MLP training tools.
- Performed SMD assembly of CoinFT PCBs using laser-cut solder-paste stencils and hot-air rework; designed a modular ESP32 breakout board for reusable prototyping.

Baskin Engineering, UC Santa Cruz **Santa Cruz, CA**

ECE 13: Computer Systems and C Programming - Group Tutor *September 2026 - Present*

- Tutor undergraduate students in C programming and embedded systems, helping students develop solutions, debug programs, and understand course concepts.

Oakland High School Robotics Club **Oakland, CA**

Lead Coder & Builder *February 2021 – May 2023*

- Designed robot assemblies and components in Onshape and fabricated custom hardware using 3D printing.
- Programmed drivetrain and end-effector systems in JavaScript and integrated mechanical and software components.
- Trained teammates to operate and maintain robotic hardware through hands-on demonstrations and walkthroughs.

PROJECTS

Mecanum Drive Train Robot

- Designed and built a mecanum-drive robot using an ESP32, four DC gearmotors with encoders, motor drivers, and 80 mm mecanum wheels.
- Selected motors, drivers, power converters, and level-shifting components based on torque, current, voltage, and encoder requirements using manufacturer datasheets.
- Designed the robot chassis and electrical layout in Onshape, fabricated the chassis using 3D printing, and assembled a LiPo-powered motor and control system.
- Prototyped motor-control and power-distribution circuits, using multimeters and voltage measurements to troubleshoot motor driver, voltage regulation, and encoder interface issues.

ECE 118 Mechatronics Project

- Led electrical development for a team-built mechatronic system, integrating DC motors, light sensors, a UC32 controller, motor drivers, and battery power.
- Designed and tuned an analog filter to pass a 2 kHz signal while achieving -42.9 dB attenuation at 1.7 kHz and 2.29 kHz, validating the response with an oscilloscope and waveform generator before soldering the final circuit onto perfboard.
- Created the system wiring diagram and fabricated mechanical components using laser cutting; prototyped a Geneva wheel mechanism for ball dispensing using an ESP32.

PIC32 Embedded Systems & Device Drivers

- Developed embedded C software for a PIC32 microcontroller, including UART, SPI, I^2C, ADC, PWM, timer, and interrupt-driven peripherals.
- Implemented a UART communication system with receive/transmit interrupts, circular buffering, packet parsing, and checksum verification for communication with a Python program.
- Developed device drivers for a rotary encoder, RC servo, and ultrasonic sensor using SPI, output compare, timers, and interrupt-based input capture.

SKILLS

Technical: Onshape, 3D Printing, C, Soldering, PCB Assembly, Oscilloscopes, Multimeters